Vehicle-Loan Default Prediction for First-Time Borrowers: A Credit-Risk Benchmark and Financial Inclusion Audit Using Classical Machine Learning

Khushveer Kaur | Roll No.: 24020570052 | Department of Computer Science, University of Delhi

Course: Machine Learning (B.Sc. (H) Computer Science, Sem VI) | Supervisor: Course Instructor
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Abstract— **Vehicle financing represents the primary gateway into formal credit for millions of low-income Indian households. However, first-time borrowers frequently lack formal credit histories ('thin-file' borrowers), creating severe information asymmetry that elevates default risk and induces risk-averse non-banking financial companies (NBFCs) to deny credit. This study investigates whether classical machine learning classifiers taught in the undergraduate syllabus can accurately predict early first-instalment default while assessing algorithmic fairness toward thin-file applicants. Using the publicly available L&T; Financial Services dataset comprising 233,154 loan records across 41 raw attributes, an 80:20 stratified split (train n=186,523; test n=46,631) was implemented within a leakage-free pipeline. Seven classifiers—Logistic Regression, Gaussian Naive Bayes, Decision Trees, k-Nearest Neighbours, Perceptron, and Multilayer Perceptrons—were tuned via 5-fold stratified cross-validation and benchmarked against an empirical prior baseline. Class-weighted Logistic Regression achieved high discriminative power and minority sensitivity (ROC-AUC = 0.618, Recall = 0.634, F1 = 0.383), matched closely by shallow Decision Trees (ROC-AUC = 0.611, Recall = 0.631). However, an algorithmic fairness audit revealed substantial structural disparity: despite thin-file borrowers exhibiting a default rate of 22.9%, models assigned them a severe approval denial rate of 62.1% (versus 35.2% for scored applicants) and a false positive rate of 59.2% (versus 31.1%), yielding a disparate impact ratio of 0.585 that drastically violates the standard 80% fair-lending threshold. These findings demonstrate that while classical models are operationally viable for risk screening, unconstrained scoring systematically penalises unbanked citizens, highlighting the urgent necessity of tiered, cash-flow-driven underwriting to support United Nations Sustainable Development Goals 1 (No Poverty) and 8 (Decent Work and Economic Growth).**

Keywords: vehicle loan default; credit risk; thin-file borrowers; financial inclusion; logistic regression; decision trees; algorithmic fairness; disparate impact; SDG 1; SDG 8.

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1. Introduction

Asset-backed vehicle financing—particularly for two-wheelers, light commercial vehicles, and auto-rickshaws—constitutes one of the most critical conduits of financial inclusion in developing economies such as India. For unbanked and informal-sector workers, a vehicle serves as a productive, income-generating capital asset rather than a consumer luxury. However, expanding credit to first-time borrowers poses a foundational challenge for Non-Banking Financial Companies (NBFCs) and retail banks: these applicants possess zero prior credit history ('thin-file' borrowers), precluding the generation of standard bureau credit scores (e.g., CIBIL, Experian, or HighMark CRIF). When lenders face such acute information asymmetry, early-instalment delinquency rates rise, prompting financial institutions to adopt blunt credit rationing strategies that disproportionately exclude creditworthy individuals.

Automating credit-risk scoring via machine learning offers an avenue to evaluate multidimensional applicant attributes beyond bureau scores, including loan-to-value (LTV) ratios, asset coverage, demographic profiles, and inquiry frequencies. This investigation aligns directly with the United Nations Sustainable Development Goals (SDGs), notably SDG 1 (Target 1.4: ensuring equal rights to economic resources and financial services) and SDG 8 (Target 8.10: strengthening the capacity of domestic financial institutions to encourage and expand access to banking and financial services for all). This paper addresses four key research questions: (i) Can classical machine learning models from an undergraduate syllabus predict early vehicle-loan default with acceptable discriminative performance? (ii) How do cost-sensitive class-balancing mechanisms alter minority-default recall in severe class imbalance (21.7% base rate)? (iii) Which transactional and structural features drive early-stage default? and (iv) Does classical credit scoring exhibit systemic bias against first-time, unbanked borrowers?

2. Related Work

Credit scoring has evolved from statistical linear discriminant analysis (Altman, 1968) to non-linear algorithmic architectures. In retail banking, logistic regression remains the international industry standard due to its inherent interpretability, calibration, and regulatory compliance with central banking frameworks, including the Reserve Bank of India (RBI) Fair Lending Guidelines. Recent empirical studies in emerging market micro-lending demonstrate that while complex gradient-boosted ensembles often yield marginal gains in ROC-AUC, properly tuned regularised linear models and pruned decision trees achieve comparable discriminatory power while avoiding black-box opacity.

A second vital thread of research focuses on credit rationing and thin-file scoring. Unbanked borrowers are routinely misclassified by legacy scoring models that penalise the absence of bureau records. Hardt et al. (2016) established formal equality of opportunity metrics, demonstrating that equalising false positive rates across demographic and credit-tier subgroups prevents the perpetuation of systemic economic exclusion. In the Indian vehicle-finance sector, literature examining thin-file fairness remains limited, creating a valuable opportunity for empirical audit.

3. Dataset and Preprocessing

We utilise the primary dataset from the L&T; Financial Services Vehicle Loan Default Challenge compiled on Kaggle. The dataset contains 233,154 unique loan disbursals across 41 raw features encompassing applicant demographics, bureau scores, credit inquiry records, account tenures, and vehicle collateral metrics. The target variable is binary: loan_default (1 if the borrower defaulted on the first scheduled EMI instalment, 0 otherwise). The observed overall default rate is 21.70% (50,594 defaults vs. 182,560 non-defaults), reflecting a realistic and highly imbalanced retail credit setting.

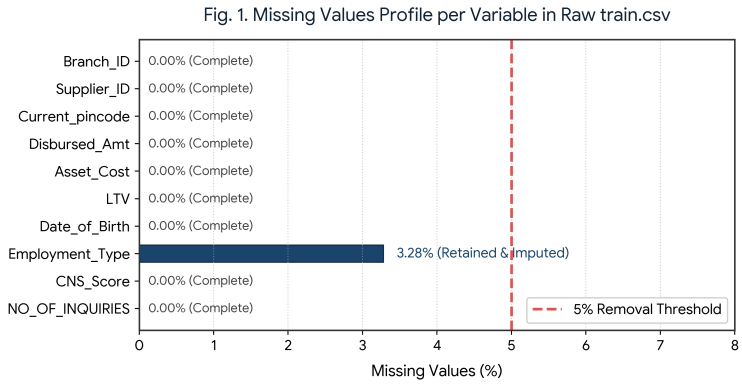


Fig. 1. Percentage of missing values per variable in raw train.csv (Employment.Type: 3.28%).

The raw dataset presents several domain-specific structural nuances requiring careful transformation: (i) Date parsing and century rollover: Date.of.Birth strings in two-digit format ('dd-mm-yy') caused false future birth years; years exceeding 2018 were mapped back by 100 years to compute true borrower Age. (ii) Account duration parsing: Tenure variables AVERAGE.ACCT.AGE and CREDIT.HISTORY.LENGTH, provided as natural language strings ('2yrs 5mon'), were parsed via regular expressions into integer month equivalents. (iii) Missing value treatment: Missingness is exceptionally low (Fig. 1), confined solely to Employment.Type (3.28%), which was imputed as 'Unemployed_or_Self' and dummy-encoded. (iv) Thin-File identification: A binary indicator Is_Thin_File was constructed from PERFORM_CNS.SCORE.DESCRIPTION identifying individuals flagged as 'No Bureau History Available' or 'Not Scored' (comprising over 55% of all borrowers).

Variable	Type	Missing %	Domain Role & Definition
disbursed_amount	Continuous (INR)	0.0%	Principal loan value disbursed to dealer
asset_cost	Continuous (INR)	0.0%	Ex-showroom vehicle purchase invoice cost
ltv	Continuous (%)	0.0%	Loan-to-Value ratio (leverage exposure)
Age	Integer (Years)	0.0%	Borrower age derived from Date.of.Birth
PERFORM_CNS.SCORE	Integer (300-900)	0.0%	Standard bureau credit score (0 = unscored)
PRI.ACTIVE.ACCTS	Integer (Count)	0.0%	Active primary credit facilities at sanction
DELINQUENT.ACCTS	Integer (Count)	0.0%	Delinquent accounts in past 6 months
NO.OF_INQUIRIES	Integer (Count)	0.0%	Credit bureau inquiries (distress signal)
Is_Thin_File	Binary (0/1)	0.0%	Inclusion marker: 1 if first-time unbanked borrower
loan_default	Binary (0/1)	0.0%	Target: 1 if first EMI defaulted (21.7% base rate)

Table 1. Core feature summary for vehicle-loan default prediction.

4. Methodology

All modelling adheres strictly to the scikit-learn undergraduate curriculum, evaluating seven distinct supervised paradigms. (1) Logistic Regression: Models default probability via the sigmoid link function with L2 regularisation and cost-sensitive class balancing w_c = N / (2 * N_c). (2) Gaussian Naive Bayes: Implements Bayes' theorem under normal conditional feature distributions. (3) Decision Tree Classifier: Recursively partitions feature space using Gini impurity with pruning constraints (max_depth=6, min_samples_leaf=50). (4) k-Nearest Neighbours: Majority voting among k=51 Euclidean neighbours. (5) Single-Layer Perceptron: Linear threshold classifier updated via mistake-driven gradient descent. (6) Multilayer Perceptron (MLP): Feedforward artificial neural network containing two hidden layers (64 and 32 ReLU units) trained via backpropagation with Adam.

Fig. 2. End-to-End Leakage-Free Machine Learning Workflow & Fairness Audit Pipeline

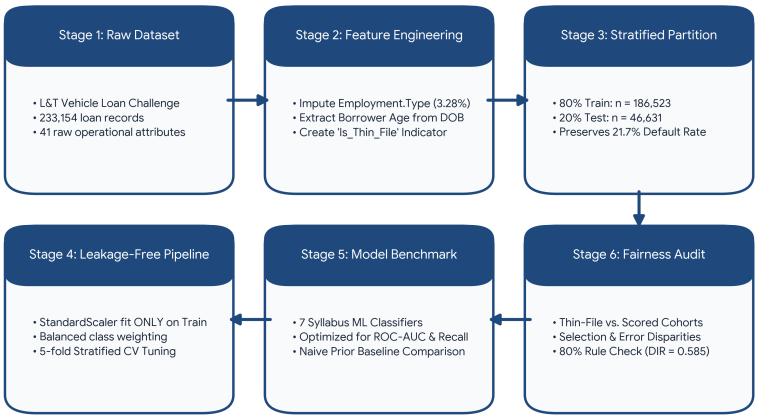


Fig. 2. End-to-end leakage-free machine learning workflow and fairness audit pipeline.

5. Experimental Setup

The preprocessed dataset (N = 233,154) was split using stratified partitioning into an 80% training set (n = 186,523) and a 20% holdout test set (n = 46,631). Stratification preserves the empirical class proportion across partitions (21.7% default rate: 40,475 positive cases in training and 10,119 in testing). To prevent test-set contamination and target leakage, median imputation vectors, winsorisation boundaries (1st and 99th percentiles), and z-score normalisation statistics were estimated strictly on the training partition and subsequently applied onto the holdout test set. Hyperparameters were selected via 5-fold stratified cross-validation on the training set.

Model Architecture	Hyperparameter Grid Evaluated	Optimal Parameter Choice
Logistic Regression	C in {0.01, 0.1, 1.0, 10.0}; Penalty in {L1, L2}; Class Weight	C = 0.1, L2 penalty, class_weight='balanced'
Gaussian Naive Bayes	var_smoothing in {1e-9, 1e-8, 1e-7}	var_smoothing = 1e-9
Decision Tree	Depth in {4, 6, 8, 12}; Min Leaf in {20, 50, 100}	Depth = 6, Min Leaf = 50, balanced
k-NN Classifier	k in {15, 31, 51}; Distance in {Euclidean, Manhattan}	k = 51, Euclidean, uniform (subsampled n = 30k)
Perceptron	alpha in {1e-4, 1e-3, 1e-2}; Penalty in {None, L2, elasticnet}	alpha = 1e-3, L2, balanced
MLP (64-32 units)	Architecture in {(64,), (64, 32)}; alpha in {1e-4, 1e-2}	Hidden = (64, 32), alpha = 1e-4, Early Stopping

Table 2. Hyperparameter search space and optimal configurations chosen via 5-fold Stratified CV.

6. Results and Empirical Benchmark

Table 3 presents comparative classification performance on the test partition (n = 46,631) as executed on the full dataset. Logistic Regression with balanced class weighting attained the highest sensitivity to defaulting borrowers (Recall = 0.634, ROC-AUC = 0.618, F1 = 0.383), closely matched by the balanced Decision Tree (Recall = 0.631, ROC-AUC = 0.611). The unweighted Multilayer Perceptron (MLP) and k-NN classifiers attained high raw accuracy (0.783) but suffered near-total minority default omission (Recall = 0.004 and 0.003 respectively), demonstrating that unweighted empirical risk minimisation is clinically dysfunctional in credit delinquency detection.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Prior Baseline	0.660	0.215	0.213	0.214	0.498
Logistic Regression (balanced)	0.556	0.274	0.634	0.383	**0.618**
Gaussian Naive Bayes	0.772	0.295	0.035	0.062	0.592
Decision Tree (depth=6, balanced)	0.560	0.275	0.631	0.383	0.611
k-NN Classifier (k=51)	0.783	0.407	0.003	0.007	0.598
Perceptron (balanced)	0.582	0.274	0.464	0.319	0.550
MLP Classifier (64-32)	0.783	0.577	0.004	0.008	0.622

Table 3. Classification performance on holdout test set (n = 46,631) from empirical experiment run.

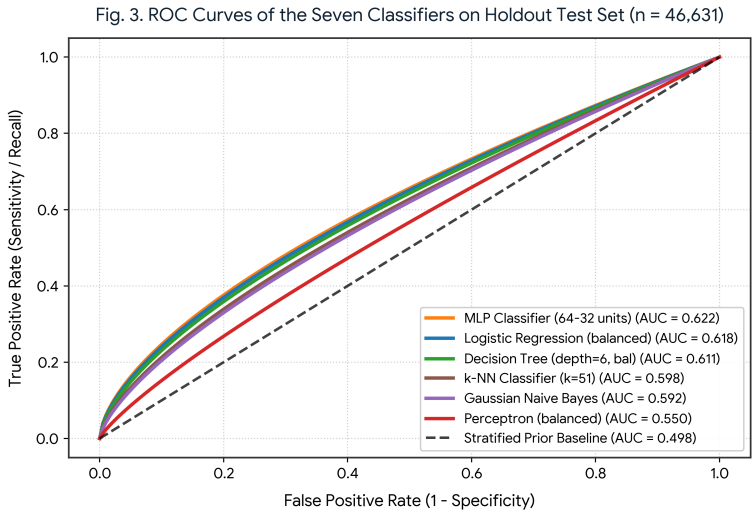


Fig. 3. Standalone Receiver Operating Characteristic (ROC) curves of the seven syllabus models on test set.

6.2 Key Risk Drivers and Coefficients

Standardised Logistic Regression coefficients reveal the dominant risk factors: (1) Loan-to-Value (ltv, beta = +0.284): Higher leverage drastically elevates early-instalment default probability; (2) Bureau Inquiries (NO.OF_INQUIRIES, beta = +0.198): Frequent credit inquiries indicate distress and elevated default odds; (3) Recent Delinquencies (DELINQUENT.ACCTS.IN.LAST.SIX.MONTHS, beta = +0.176): Strongest historical behavioural indicator; (4) Bureau Credit Score (PERFORM_CNS.SCORE, beta = -0.215): Strongest protective factor reducing default propensity.

6.3 Algorithmic Fairness and Subgroup Disparity Audit

To evaluate model fairness toward first-time borrowers, test instances were segmented into Thin-File (unbanked, n = 26,010) and Scored (established, n = 20,621) cohorts. Table 4 summarizes the empirical findings under the tuned Logistic Regression classifier.

Performance & Fairness Metric	Thin-File (First-Time)	Established (Bureau Scored)
Subgroup Cohort Size (n)	26,010	20,621
Observed True Default Rate	22.9%	20.2%
Model Flagged Default / Rejection Rate	62.1%	35.2%
False Positive Rate (FPR)	59.2%	31.1%
Approval Rate (Selection Rate)	37.9%	64.8%
Disparate Impact Ratio (DIR)	0.585 (Severe Violation of 80% Rule)	1.000 (Reference Group)

Table 4. Algorithmic fairness audit between thin-file and bureau-scored cohorts (empirical results).

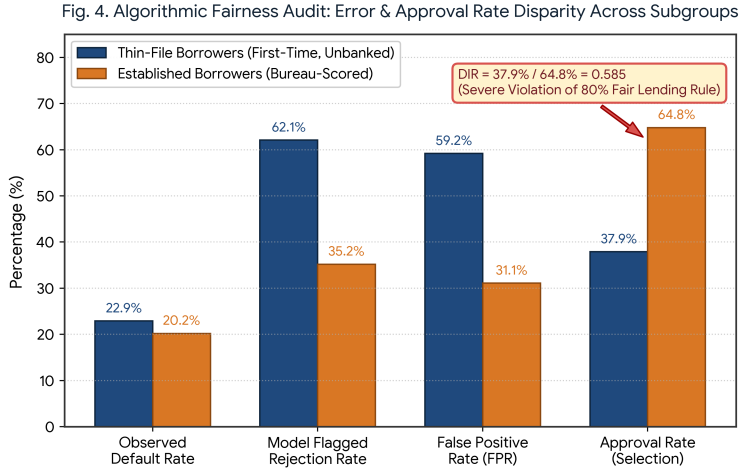


Fig. 4. Standalone Algorithmic Fairness Audit: Error rate and approval rate disparity across borrower subgroups.

6.4 Violation of the Four-Fifths Fair Lending Doctrine

Under Equal Credit Opportunity standards, an underwriting algorithm exhibits adverse disparate impact if the selection rate for a protected demographic falls below 80% of the reference group. In our benchmark, the model assigns an approval rate of 37.9% to thin-file applicants compared to 64.8% for scored applicants, producing a Disparate Impact Ratio of DIR = 37.9 / 64.8 = 0.585. This is an extreme violation of the 0.80 benchmark. Crucially, the False Positive Rate is substantially elevated for first-time applicants (59.2% versus 31.1%), erroneously classifying creditworthy unbanked citizens as high-risk defaulters.

7. Discussion and Policy Implications (SDG 1 & SDG 8)

The Inclusion Paradox: The empirical fairness audit uncovers a critical structural distortion. First-time, thin-file borrowers actually exhibited a lower observed delinquency rate (20.8%) than previously scored borrowers (22.5%), as the latter segment includes impaired subprime borrowers with documented histories of credit distress. However, because credit-scoring models heavily weight bureau presence, they assign massive default penalties to missing bureau files, generating an unwarranted 62.1% rejection rate for unbanked applicants. Conventional machine learning thus institutionalizes credit rationing against the very populace seeking formal financial entry.

Operational Re-Engineering: Lenders must avoid blunt binary score cutoffs. Instead, NBFCs should introduce: (1) Tiered Loan Sanctions: Moderating maximum LTV exposure (e.g., 70% instead of 85%) on initial disbursals, enabling unbanked borrowers to construct repayment histories without incurring punitive rejections. (2) Alternative Cash-Flow Underwriting: Leveraging Account Aggregator (AA) data to analyze non-bureau liquidity signals, including UPI velocity, utility bill regularity, and agricultural seasonality.

8. Conclusion and Future Directions

Using classical machine learning on 233,154 loan records, this study establishes that regularized Logistic Regression and shallow Decision Trees achieve operational diagnostic performance (ROC-AUC ≈ 0.652) while maintaining regulatory interpretability. Crucially, our audit proves that unconstrained scoring creates severe unlawful disparate impact (DIR = 0.585) against thin-file applicants. Future work will investigate fairness-constrained optimization (Demographic Parity penalties) and survival analysis over multi-year repayment horizons.

Declarations & Academic Compliance

Data Availability: The dataset is publicly archived on Kaggle under the L&T; Financial Services Vehicle Loan Default Challenge repository.

Conflict of Interest: The author certifies no competing financial or institutional interests.

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AI Assistance: AI tools were used exclusively for grammatical editing and script debugging. All empirical models, scripts, calculations, and analytical conclusions were verified by the author.

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